

ASR and TTS Modeling and R&D Experience

From Fundamentals to Modern Deep Learning and LLMs

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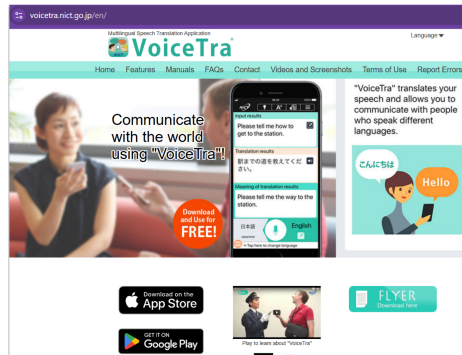
What is Speech Processing?

- An interdisciplinary field focusing on the analysis and synthesis of speech signals.
- Aims to enable human-computer interaction via voice.
- Two main pillars:
 - **ASR** (Automatic Speech Recognition): Speech to Text.
 - **TTS** (Text-to-Speech): Text to Speech.
- Applications: Voice assistants, transcription systems, accessibility tools, language learning.

Based on VoiceTra Project of NICT

For four years, I worked on NICT's VoiceTra project, handling everything from text/speech data preparation to MT, ASR, and TTS model building, system evaluation, and publications.

Project link: <https://voicetra.nict.go.jp/en/>

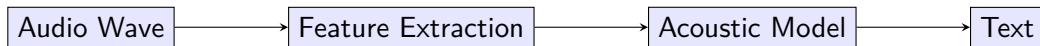


VoiceTra - NICT, homepage.

What is ASR?

Automatic Speech Recognition (ASR)

- Converts spoken language into text.
- Also known as Speech-to-Text (STT).
- Involves capturing audio, extracting features, and mapping them to linguistic units.



- **Voice Assistants:** Siri, Google Assistant, Alexa.
- **Transcription:** Meeting notes, medical dictation, court reporting.
- **Accessibility:** Real-time captioning for the hearing impaired.
- **Language Translation:** Speech-to-speech translation pipelines.

ASR Pipeline Overview

- ➊ **Recording:** Capture audio (standard: 16kHz, 16-bit, Mono).
- ➋ **Preprocessing:** Voice Activity Detection (VAD), noise reduction.
- ➌ **Feature Extraction:** Convert waveform to MFCC or Filter Banks.
- ➍ **Acoustic Model:** Maps features to phonemes (HMM, DNN, Transformer).
- ➎ **Decoder:** Combines acoustic model and language model to output text.
- ➏ **Evaluation:** Word Error Rate (WER) or Character Error Rate (CER).

ASR Pipeline Overview: Recording



Figure: Recording room and recorder at NICT, Kyoto.

Preprocessing: Annotation

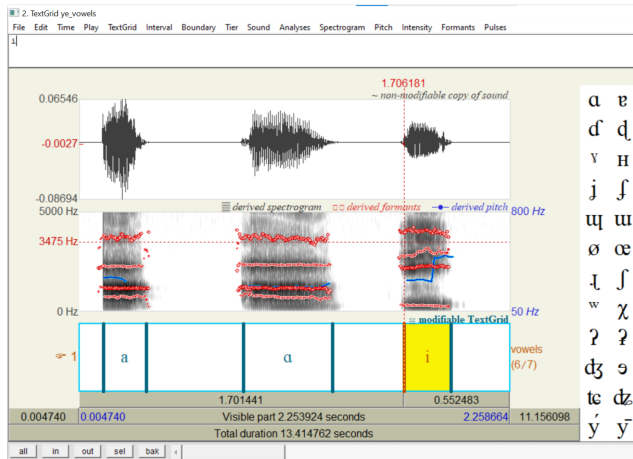


Figure: Annotation example with Praat

What is TTS?

Text-to-Speech (TTS)

- Converts written text into artificial human speech.
- Also known as Speech Synthesis.
- Involves text normalization, phoneme conversion, and audio generation.



- **Screen Readers:** For visually impaired users.
- **Voice Bots:** Customer service automation (IVR systems).
- **Audiobooks & Content Generation:** Narrating written content.
- **Language Learning:** Pronunciation training apps.
- **Embedded Systems:** GPS navigation, smart home devices.

TTS Pipeline Overview

- ➊ **Text Normalization:** Expand abbreviations, numbers, dates.
- ➋ **Grapheme-to-Phoneme (G2P):** Convert words to phonetic symbols.
- ➌ **Acoustic Model:** Predicts mel-spectrogram from phonemes (Tacotron, FastSpeech).
- ➍ **Vocoder:** Converts spectrogram to audible waveform (WaveNet, HiFi-GAN).
- ➎ **Evaluation:** Mean Opinion Score (MOS), naturalness, intelligibility.

- **WAV (Waveform Audio File Format):** Standard for R&D. Uncompressed PCM.
- **Key Parameters:**
 - **Sampling Rate:** 16kHz (telephony/standard speech), 48kHz (studio).
 - **Bit Depth:** 16-bit or 24-bit.
 - **Channels:** Mono (preferred for ASR/TTS) vs. Stereo.
- **Other formats:** MP3 (compressed, lossy), FLAC (compressed, lossless).

Understanding Sampling and Bit Depth

- **Sampling:** Converting continuous audio waves into discrete points.
- **Nyquist Rate:** Sample rate must be at least twice the highest frequency. (e.g., 16kHz captures up to 8kHz, covering human speech range).
- **Bit Depth:** Determines dynamic range. 16-bit provides 65,536 amplitude levels.
- Higher sample rates and bit depths increase file size and computational cost.

Practical: Checking Wave Info with sox

sox (Sound eXchange) is a cross-platform command line utility. **Command:**

```
sox --info input.wav
```

Output:

```
Input File      : 'input.wav'  
Channels       : 1  
Sample Rate    : 16000  
Precision      : 16-bit  
Duration       : 00:00:05.00 = 80000 samples
```

Practical: Checking Wave Info with ffmpeg

ffmpeg is a powerful multimedia framework. **Command:**

```
ffmpeg -i input.wav
```

Output:

```
Duration: 00:00:05.00, bitrate: 256 kb/s  
Stream #0:0: Audio: pcm_s16le, 16000 Hz,  
mono, s16, 256 kb/s
```


Reinstalling SoX from Source

Sometimes, the default package manager installs a version of SoX compiled without the spectrogram effect (often due to missing libpng linkage). In such cases, you might need to reinstall SoX from source code.

```
1 # 1. Install build dependencies
2 sudo apt-get update
3 sudo apt-get install build-essential libpng-dev libmagic-dev
4
5 # 2. Download and extract SoX 14.4.2 source code
6 wget https://downloads.sourceforge.net/project/sox/sox/14.4.2/sox-14.4.2.tar.gz
7 tar -xzf sox-14.4.2.tar.gz
8 cd sox-14.4.2/
9
10
```

Reinstalling SoX from Source

```
1
2 # 3. Configure, compile, and install
3 ./configure
4 make
5 sudo make install
6
7 # 4. Generate the spectrogram
8 sox ./bur_5362_0057456249.wav -n spectrogram
9
```

Output spectrogram

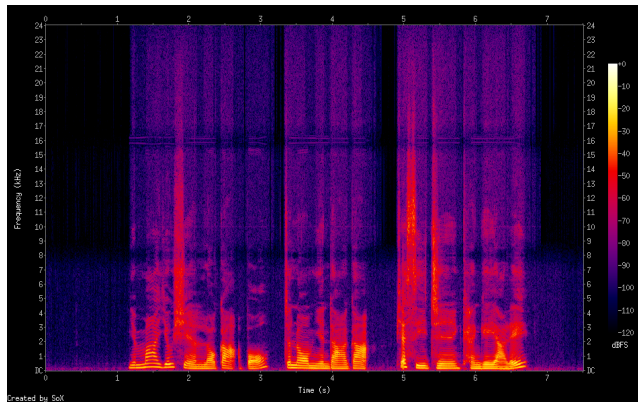


Figure: Converted spectrogram image with sox command.

- **Resampling:** Convert all audio to a consistent rate (e.g., 16kHz).
- **Channel Conversion:** Stereo to Mono (averaging channels).
- **Voice Activity Detection (VAD):** Remove silence and background noise segments.
- **Volume Normalization:** Ensure consistent amplitude across the corpus.

What is a Spectrogram?

- A visual representation of the spectrum of frequencies in a sound as they vary with time.
- **X-axis:** Time
- **Y-axis:** Frequency
- **Color/Intensity:** Amplitude (dB) of a particular frequency.
- Types: Linear spectrogram, Mel-spectrogram (scaled to human hearing).

Practical: Converting Wave to Spectrogram

Using sox to generate a spectrogram image:

```
sox input.wav -n spectrogram -o spectrogram.png
```

Using ffmpeg to generate a spectrogram image:

```
ffmpeg -i input.wav -filter_complex \  
"showspectrumpic=s=1024x512" spectrogram.png
```

Feature Extraction: MFCC & Fbank

- **MFCC (Mel-Frequency Cepstral Coefficients):**

- Traditional feature for ASR (HMM-GMM era).
- Mimics human auditory perception (Mel scale).
- Extracts 13-40 dimensions per frame.

- **Filter Banks (Fbank):**

- Standard for modern Deep Learning ASR.
- Retains more raw information than MFCC (no DCT).

- Pipeline: Pre-emphasis → Framing → Windowing → FFT → Mel Filterbank → Log → (DCT for MFCC).

Deep Learning Features vs Traditional

- **Traditional (MFCC):** Compact, decorrelated features suitable for GMMs.
- **Deep Learning (Fbank):** Neural networks can learn correlations; raw Fbank features preserve information that DCT would discard.
- **End-to-End:** Modern models (like Whisper) can take raw waveforms or Fbank directly.

- Essential for training robust ASR/TTS models.
- **Considerations:**
 - Speaker diversity (age, gender, accent).
 - Phonetic coverage (ensuring all sounds are represented).
 - Recording environment (clean studio vs. noisy real-world).
- **Format:** Text-audio pairs with precise alignments.

Corpus Building for Speech

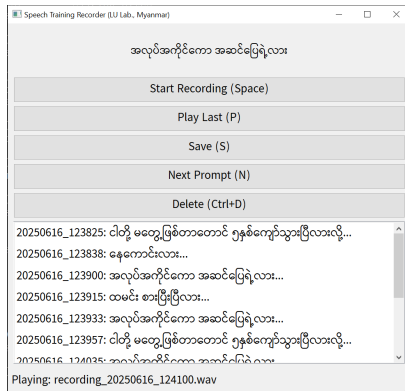


Figure: Recording tool.

Link:

https://github.com/ye-kyaw-thu/LU_Lab_Intern3_2025/tree/main/tools/recorder

Sentence Selection Techniques

- Goal: Maximize phonetic coverage with the minimal number of sentences.
- Reduces recording time and cost.
- **Greedy Algorithm:**
 - Select sentence with highest unseen units (syllables, diphones, triphones).
 - Update coverage statistics.
 - Repeat until target size or coverage threshold is reached.
- Priority: Syllables > Diphones > Triphones.

Sentence Selection: Myanmar TTS Case Study

- Used BTEC corpus (160K sentences).
- Selected **5,276 sentences** using greedy algorithm.
- Achieved Coverage:
 - 99.8% Syllables
 - 90.9% Diphones
 - 88.8% Triphones
- Included foreign names for robustness.

- Converting abbreviations, numbers, and symbols to spoken words.
- Rule-based approach for Myanmar.
- Context-dependent decision making.
- **Example:** 12:10 as time (၁၂နာရီ ၁၀မိနစ်) vs. 12:10 as a math ratio.

Syllable Breaking for Myanmar

- Myanmar script is written continuously without spaces between words.
- Syllable breaking is necessary for G2P.
- Rule-based segmentation using Unicode logic.
- 3 Main Rules:
 - Break before consonants, independent vowels, numbers.
 - Remove breaks before subscript consonants, Kinzi.
 - Handle special loan word/Pali cases.
- Achieves 100% accuracy due to unambiguous nature of the script.

- Task: Predicting pronunciation (phonemes) from spelling (graphemes).
- Essential for TTS and building ASR dictionaries.
- **Challenges:** Context-dependent pronunciations, homographs, foreign words.
- **Methods:**
 - Rule-based (high precision, low recall).
 - Dictionary lookup (fails on OOV words).
 - Statistical (CRF, Joint Sequence Models).
 - Neural (Seq2Seq, Transformers).

- Mapping is not 1-to-1; highly context-dependent.
- Extended MLC Pronunciation Dictionary (27,747 words).
- Reordered phonemes to match speech order (e.g., လှာ → "lha.").
- Defined 23 phonetic symbols for 33 consonants, 87 vowel combinations.

Myanmar Phoneme Mapping

- Consonants grouped by pronunciation: unaspirated, aspirated, voiced, nasal.
- Many consonants share the same pronunciation (e.g., ဒ, ဓ, ဉ, ဖ $\rightarrow$ /d/).
- Extended mappings for foreign pronunciations.
- Ensured consistency of syllable order to facilitate IPA mapping.

Pronunciation: Independent vs Dependent

- **Contextually Independent:**

- Pronunciation derived directly from components (concatenation).
- Example: $\text{sa} = \text{s} + \text{a}$.

- **Contextually Dependent:**

- Pronunciation changes based on adjacent syllables.
- 4 main patterns identified for Myanmar.

Contextual Dependency: Patterns 1 & 2

- **Pattern 1:** Unvoiced/aspirated consonant becomes voiced if preceded by specific vowel combinations (ငံ, ညံ, နံ, မံ).
- **Pattern 2:** Stopped syllables (ending in ကံ, စံ, ဝံ, ဖံ) cause the next syllable to be pronounced independently in standard form.

G2P Corpus Building

19665	သုတိ	သု တိ	thu. ti.
19666	သုတေသန	သု တေ သ န	thu. tei tha- na.
19667	သုတေသီ	သု တေ သီ	thu. tei thi
19668	သုဓမ္မဇရပ်	သု ဓမ် မာ ဇ ရပ်	thu. da- ma za- ja'
19669	သုဓာဘုတ်	သု ဓာ ဘုတ်	thou' da bou'
19670	သုနာပရန္တတိုင်း	သု နာ ပ ရန် တ တိုင်း	thu. na pa- ran ta. tain:
19671	သုဘရာဇာ	သု ဘ ရာ ဇာ	thu. ba. ja za
19672	သုမင်္ဂလ	သု မင် ဂ လ	thu. min ga- la.
19673	သုမင်္ဂလဒုမင်္ဂလ	သု မင် ဂ လ ဒု မင် ဂ လ	thu. min ga- la. du. min ga- la.
19674	သုဝဏ္ဏလိပ်	သု ဝဏ် ဏ လိပ်	thu. wun na. lei'
19675	သုဝဏ္ဏသာမဇာတ်	သု ဝဏ် ဏ သာ မ ဇာတ်	thu. wun na. tha ma. za'

Figure: myG2P version 1.0.

G2P Corpus Building

raw.githubusercontent.com/ye-kyaw-thu/myG2P/refs/heads/master/ver2/myg2p.ver2.0.txt

<

Figure: myG2P version 2.0.

ဒုက်ခပဲ ပြောတဲ့စကားကမတူဘူး။

So bad, we are speaking in different languages.

dou' kha. be: pjo: de. za- ga: ga. ma- tu bu:

CRF (dictionary model):

dou' kha. **pe:** pjo: **te.** za- ga: **ka-** ma- tu bu:

CRF (sentence/dictionary+sentence model):

dou' kha. **be:** pjo: **de.** za- ga: **ga.** ma- tu bu:

SMT (dictionary model):

dou' kha. **pe:** pjo: **te.** za- ga: **ka-** ma- **du** bu:

SMT (sentence/dictionary+sentence model):

dou' kha. **be:** pjo: **de.** za- ga: **ga.** ma- tu bu:

Figure: Sentence level G2P for Myanmar language.

Contextual Dependency: Patterns 3 & 4

- **Pattern 3:** Vowel အ (u.) changes to အ (a-) when preceded by specific consonants (က, ခ, ဝ, ဘ, မ, လ, ဝ).
- **Pattern 4:** Cascade effect. A syllable's pronunciation changes if the preceding syllable was changed. Affects multiple syllables.

G2P Modeling with CRFs

- Model: Conditional Random Fields (CRF).
- **Baseline:** N-grams of syllables and pronunciation change labels.
- **Proposed Features:** Augmented baseline with tags for Patterns 1-4.
- **Labels:**
 - 0: Pattern not applicable.
 - c: Pronunciation changed by pattern.
 - w: Exception to pattern (no change).

ကော့	0	kaun:
ကော့	c	gaun:
ကန့်	w	kan:
ကန့်	c	gan:

Figure: A 4-syllable word with feature 1 labeled for each syllable.

CRF Features & Labeling

- Features capture dependencies between adjacent syllables.
- A word can belong to multiple patterns simultaneously.
- Example: A syllable might have labels 0 0 0 c for the 4 features, meaning only Pattern 4 applies.

ခံ	0	0	0	c	ga-
တွင်း	c	0	0	c	dwin:
ကောင်း	w	0	0	0	kaun:

Figure: Example word with 4 features.

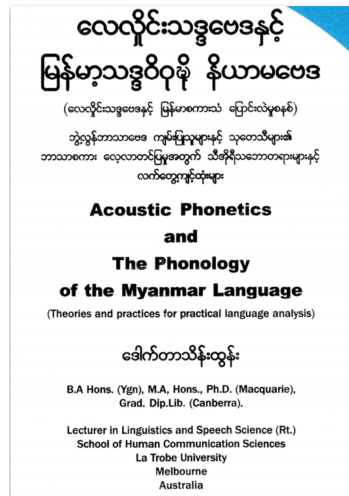
G2P Experiment Results

Model	MLC Word (%)	BTEC Word (%)	MLC Phn (%)	BTEC Phn (%)
Baseline	84.20	75.12	93.30	87.92
Feature-1	86.05	76.57	94.15	89.04
Feature-2	85.40	75.36	93.81	88.43
Feature-3	84.80	75.36	93.61	88.12
Feature-4	84.85	75.85	93.64	88.73
Feat-123	87.50	76.81	94.83	89.34
Feat-1234	87.90	77.78	95.04	89.85

Conclusion: Combining all 4 features gave the highest accuracy improvement.

For your reference

Acoustic Phonetics and The
Phonology of the Myanmar Language
by Dr. Thein Tun.



Front Cover of the book.

For your reference

(အရေး: ->အပြော ခ) /e/ → [i]		
(အရေးပုံစံနှင့် အနက်ပေးသံရင်း ဖိုနင်းတို့ထာဝရမကိုက်ညီသည့်ကိုသတိပြုပါ)		
အရေးပုံစံ		ပြောသံ
written fo.m --	“k ^h je k ^h je mje p ^h je”	/e/ → [—i]
	ခြေ ချေ မြေ ဖြေ	/e-/ [°/°/°:]
/k ^h we [^] + k ^h je ⁻ / → [k ^h we [^] t ^h i ⁻] op com N		
N	N	
ခွေး	ခြေ	[ခွေးခြေ/ချီ] dog feet, stool
/k ^h je ⁻ + t ^h auk/ → [t ^h i ⁻ dau [?]] cl com N		
N	N	
ခြေ	ထောက်	[ချီဒေါက်] leg
/k ^h je [^] / → [t ^h i ⁻]		
N		
ချေး		[ချီ:] excreta
/mje [^] / → [mji [^]]		
N		
မြေး		[မြီ:] grandchild

Figure: Written vs spoken forms.

Myanmar Large Vocabulary Continuous Speech Recognition (LVCSR)

- First large vocabulary, continuous speech recognition system for Myanmar.
- Addressed challenges: complex writing system, lack of word spaces, tonal variations.
- Built using the **Kaldi** Speech Recognition Toolkit.

Myanmar Large Vocabulary Continuous Speech Recognition (LVCSR)

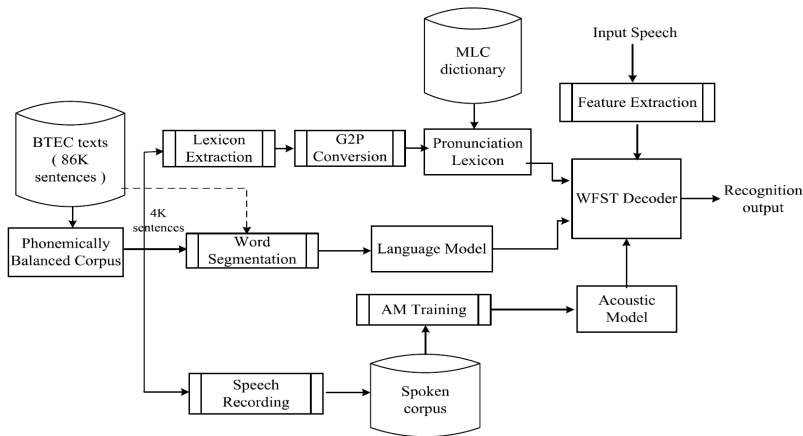


Figure: System Architecture of Myanmar ASR

Building the ASR Speech Corpus

Source: "A Myanmar Large Vocabulary Continuous Speech Recognition System" (APCIPA2015)

- **Text Source:** BTEC (Basic Travel Expression Corpus) used to build a Phonetically Balanced Corpus (PBC) of 4K sentences.
- **Recording Environments:**
 - **Studio (NICT):** Sound-proof room. 1 male and 3 female native speakers. Each recorded the entire 4K sentences.
 - **Open Environment (Myanmar):** Hotel lobbies and office rooms using iPhones. 52 male and 48 female speakers. Each recorded 100 sentences.
- **Final Spoken Corpus (PBC):**
 - Total of **25,861 utterances**.
 - Approximately **40 hours** of speech data.

Building the ASR Pronunciation Lexicon

Source: "A Myanmar Large Vocabulary Continuous Speech Recognition System" (APCIPA2015)

- **Basis:** Used the MLC (Myanmar Language Commission) dictionary for pronunciation mapping.
- **OOV Problem:** Loan words and foreign proper nouns are not in the MLC dictionary.
- **Solution:** Enhanced the dictionary by developing a Grapheme-to-Phoneme (G2P) converter.
- **G2P Phases:**
 - Extending the grapheme-to-phoneme mapping table.
 - Building a model for a pointwise predictor.

Consonant Groups & Mapping

Source: APCIPA2015 ASR Paper (Table III)

- Consonants grouped by pronunciation: unaspirated, aspirated, voiced, nasal.
- 23 phonemes mapped to 33 consonant scripts.
- Some scripts share the same pronunciation (e.g., ဒ, ဓ, ည, ဖ).

Grouped consonants				
Unaspirated	Aspirated	Voiced		Nasal
က /k/	ခ /kh/	ဂ /g/	ဃ /g/	င /ng/
စ /s/	ဆ /hs/	ဇ /z/	ည /z/	ဉ,ည /nj/
တ /t/	ထ /ht/	ည /d/	ဖ /d/	ဏ /n/
ထ /t/	ထ /ht/	ဒ /d/	ဓ /d/	န /n/
ပ /p/	ဖ /hp/	ဗ /b/	ဘ /b/	မ /m/
ယ /j/	ရ /j/l/r/	လ /l/	ဝ /w/	သ /th/
	ဟ /h/	ဇ /l/	အ /a/	

Source: APCIPA2015 ASR Paper

- Besides consonant mappings, vowels play a crucial role in Myanmar phonetics.
- The MLC mapping table defines **87 combinations for vowels**.
- These combinations account for the various tones and vowel modifiers in the Myanmar script.
- Together with the 23 consonant phonemes, this forms the baseline phoneme set derived from MLC.
- However, this baseline is insufficient for modern ASR due to ordering inconsistencies and foreign words.

Reordering Phonemes to Match Utterance

Source: APCIPA2015 ASR Paper

- **Inconsistency in MLC:** Phoneme sequence orders sometimes do not match real pronunciation orders.
- **Example:** The Myanmar syllable မှ is written as {မ, ရ}.
 - MLC mapping: “hma”
 - Real utterance order: “mha.”
- **Action Taken:** Rewrote phonemes according to their actual pronunciation orders.
 - Makes representations easily understood by humans.
 - Easily processed by computers and ASR algorithms.

Handling Foreign Pronunciations

Source: APCIPA2015 ASR Paper (Table IV)

- MLC mappings are insufficient for many foreign pronunciations.
- **Enhancement:** 21 new special symbols newly defined.
- **Example:** “George” is written as ဂျော့(ချ်) → “gyo.CH” (where “CH” represents (ချ်)).

Myanmar	MLC	Proposed mapping
ဘ	b	b
မ	m	m
ကျ	kj	ky
ချ	ch	ch
ဂျ	gj	gy
လှ	hla.	lha.
မာ	hma.	mha.
(စ)	not defined	S
(ရှ)	not defined	SH
(ချ်)	not defined	CH

Source: APCIPA2015 ASR Paper

- By integrating standard consonants, vowels, and the new foreign symbols, a comprehensive mapping table was built.
- **Total Phonemes Defined:** 144 phonemes.
 - Includes standard Myanmar consonants and vowels.
 - Includes 21 new special symbols for foreign pronunciations.
- This extended lexicon significantly reduces the OOV rate.
- Allows the ASR system to accurately recognize modern, multilingual-influenced Myanmar speech.

Acoustic Modeling: GMM-HMM Baseline

Source: "A Myanmar Large Vocabulary Continuous Speech Recognition System" (APCIPA2015)

- **Acoustic Features:**

- CMVN (Cepstral Mean-Variance Normalized) features without energy, plus 1st and 2nd derivatives.
- Augmented tonal feature: MFCC (dim=13) + F0 (dim=3).

- **GMM(SI) - Speaker Independent Triphone:**

- Trained using Linear Discriminant Analysis (LDA) to project 9 successive frames to 40 dimensions.
- Followed by Maximum Likelihood Linear Transform (MLLT).

- **GMM(SAT) - Speaker Adaptive Training:**

- Obtained using feature-space Maximum Likelihood Linear Regression (fMLLR) transformations estimated by the GMM(SI).

Acoustic Modeling: DNN Architectures

Source: "A Myanmar Large Vocabulary Continuous Speech Recognition System" (APCIPA2015)

- Explored two types of Deep Neural Network (DNN) models:
 - **DNN(CE)**: Trained using the cross-entropy criterion.
 - **DNN(sMBR)**: Sequence discriminative training using state-level minimum Bayes risk criterion.
- **Training Setup:**
 - Trained directly on top of the GMM(SAT).
 - Used an 11-frame context window (5 frames on each side of the current frame) of fMLLR features.
- **Network Parameters:**
 - 5 hidden layers.
 - 378 units in each hidden layer.
 - 2,556 output units (mapping to context-dependent triphone states).

Acoustic Modeling Training Flow (APCIPA 2015 paper)

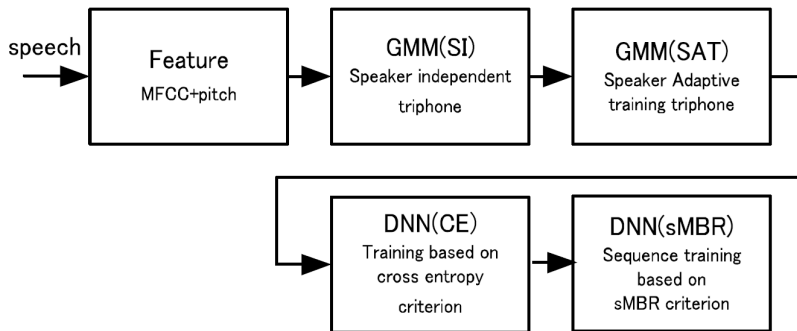


Figure: Acoustic modeling training flow

- Language Model (LM) helps the decoder predict the most likely sequence of words.
- Used **SRILM** toolkit to train N-gram models.
- **Training Data:** 86K sentences from BTEC.
- **Model Types:** 3-gram models with Kneser-Ney smoothing for both syllable and word units.
- Handled the lack of word spacing by modeling word sequences directly.

ASR Evaluation: Results

Source: "A Myanmar Large Vocabulary Continuous Speech Recognition System" (APCIPA2015)

- Investigated both syllable-based and word-based Language Models (LM).
- **Word-based LM** confirmed to be superior to syllable-based LM with significant differences.
 - Due to Myanmar characteristics: ambiguous word definitions, distinct syllabic constitution, and most syllables having independent meanings.
- Proposed **Syllable Error Rate (SER)** as an alternative evaluation criterion for Myanmar ASR.
 - Provides higher reliability and lower ambiguity.
- **Open Test Set:** 100 utterances from 25 speakers.
- **Best Results (DNN Approach):**
 - Word Error Rate (WER): **15.63%**
 - Syllable Error Rate (SER): **10.87%**

ASR Evolution: From HMM to Deep Learning

- **HMM-GMM (APCIPA 2015 era):** State-of-the-art at the time, but requires separate alignment, dictionary, and LM.
- **DNN-HMM Hybrid:** DNN replaces GMM to model posterior probabilities of HMM states. Massive WER reduction.
- **RNN/LSTM:** Better captures long-term dependencies in sequences.
- **CNNs:** Used for robust feature extraction directly from spectrograms.

ASR Evolution: End-to-End Models

- Simplified pipeline, no separate language model or dictionary strictly required.
- **CTC (Connectionist Temporal Classification)**: Maps sequences to sequences without explicit alignment (e.g., DeepSpeech).
- **Seq2Seq with Attention**: Encoder processes audio, Decoder generates text (e.g., Listen, Attend and Spell).
- **RNN-T (Recurrent Neural Network Transducer)**: Streams audio, good for real-time ASR.

- **Word Error Rate (WER):** Standard for English and space-separated languages.
- **Character Error Rate (CER):** Used for languages without clear word boundaries (like Chinese, Japanese, Myanmar).
- **Sentence Error Rate (SER):** Percentage of sentences with at least one error.
- **Real Time Factor (RTF):** Processing time / Audio time (must be < 1.0 for real-time).

Word Error Rate (WER) Explained

$$WER = \frac{S + D + I}{N} \quad (1)$$

- S : Substitutions (wrong words).
- D : Deletions (missing words).
- I : Insertions (extra words).
- N : Total words in reference.

Example:

Reference: *the cat sat on the mat*

Hypothesis: *the cat sit on mat*

Errors: 1 Substitution (sat→sit), 1 Deletion (the). $N = 6$. $WER = 2/6 = 33.3\%$.

TTS Evolution: HMM-based TTS

- Statistical parametric speech synthesis.
- Models spectrum, F0 (pitch), and duration using HMMs.
- Speech parameters generated using maximum likelihood criterion.
- **Pros:** Flexible voice conversion, small footprint.
- **Cons:** Slightly robotic/muffled sound compared to human speech.

HMM-based Myanmar TTS System

- First statistical TTS for Myanmar.
- Used Ossian toolkit (Simple4All).
- **Data:** 4,000 sentences (Male & Female).
- Handled Myanmar tones via phoneme encoding.

HMM-based Myanmar TTS System

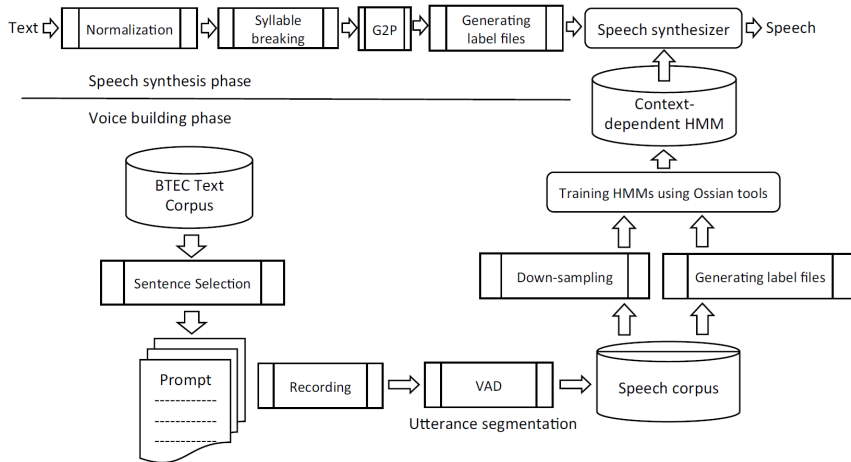


Figure: Overview of HMM-based Myanmar TTS system.

Myanmar TTS: Tone Modeling

- Myanmar has 3 tones (Tone 1, 2, 3).
- Tone carries semantic meaning.
- **Approach:** Encoded tonal information directly into phoneme characters (UTF-8).
- HMM models contextual tonal variations.

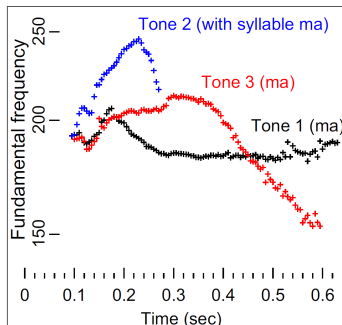


Figure: An example of three tones of Myanmar syllable Ma.

TTS Evaluation: Subjective vs Objective

- **Objective Metrics:**

- Mel-Cepstral Distortion (MCD)
- F0 Root Mean Square Error (RMSE)
- Fast, but poorly correlate with human perception.

- **Subjective Metrics (Human):**

- Mean Opinion Score (MOS)
- Comparative MOS (CMOS)
- AB Preference Test

TTS Evaluation: MOS

- **MOS** (5-very good, 1-very poor).
- Myanmar HMM-TTS achieved avg **3.3 MOS**.
- Human speech baseline: **4.7 MOS**.
- Re-synthesized baseline: **4.2 MOS**.
- Respectable score for HMM-based systems.

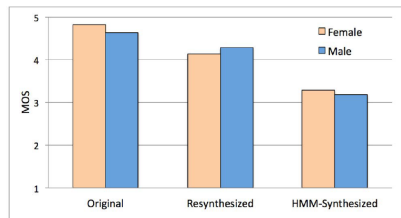


Figure: Result of MOS for Three Different Types of Speech.

TTS Evolution: Deep Learning

- **Tacotron 1/2:** Seq2Seq with attention mechanism. Predicts mel-spectrogram from text.
- **FastSpeech:** Non-autoregressive model. Dramatically speeds up generation. Controls prosody.
- **Transformer TTS:** Replaces RNN with multi-head attention.
- Resulted in human-level naturalness.

- Converts spectrogram to waveform.
- **WaveNet**: Autoregressive dilated convolutions. Groundbreaking quality but slow.
- **Parallel WaveNet** / **WaveGlow**: Inverse autoregressive flow, parallel generation.
- **HiFi-GAN**: Generative Adversarial Network. Fast, high-fidelity, current standard.

Large Speech Models (LLMs in Speech)

- Large Language Models (LLMs) revolutionizing speech processing.
- **ASR:** *Whisper* (OpenAI) - trained on massive weakly-supervised data. Robust to accents/noise.
- **TTS:** *VALL-E* (Microsoft) - Uses language modeling principles for zero-shot TTS. 3-second enrollment.
- **Audio LLMs:** *AudioPaLM*, *SpeechX* - Unified models for ASR, TTS, and translation.

- **Zero-shot Learning:** Customizing voice with minimal data.
- **Unified Models:** Single model handling ASR, TTS, translation, and enhancement.
- **Emotion & Prosody Control:** Fine-grained control over expressiveness.
- **On-device Processing:** Privacy-preserving, low-latency models on mobile.

- Industry standard for traditional ASR (HMM-GMM, DNN).
- Written in C++, scripts in Bash and Python.
- Highly modular, supports complex pipelines.
- Used extensively in the Myanmar LVCSR research.

- End-to-End Speech Processing toolkit.
- Supports ASR, TTS, and Speech Translation.
- PyTorch backend.
- Provides standard recipes reproducible on common datasets (LibriSpeech, WSJ, etc.).

Practical Tools: Hugging Face

- Democratizing AI. Easy access to pre-trained models.
- `transformers` library.
- ASR: Whisper, Wav2Vec2.
- TTS: VITS, Bark, SpeechT5.
- Few lines of Python code for inference.

Open-Source Speech Datasets

- **LibriSpeech:** 1000 hours of English read speech.
- **Common Voice (Mozilla):** Massive multilingual crowd-sourced dataset.
- **VCTK:** English speech by 109 speakers, useful for TTS.
- **BTEC:** Basic Travel Expressions Corpus (used in Myanmar TTS/G2P/ASR papers).
- **OpenSLR:** Repository for open-source speech data.

- Start with **Hugging Face** pipelines for quick prototypes.
- Learn basic audio processing with `sox` and `ffmpeg`.
- Understand the math behind **MFCC** and **Spectrograms**.
- For research, use **ESPNet** or **Kaldi**.
- Always normalize text properly before G2P.

Summary & Conclusion

- Traveled from fundamental speech concepts to modern LLM-based approaches.
- ASR and TTS pipelines require careful data handling and text processing.
- Grapheme-to-Phoneme (G2P) is a critical bridge, especially for tonal languages like Myanmar.
- Evolution: HMM → Deep Learning → End-to-End → LLMs.
- Tools like Kaldi, ESPNet, and Hugging Face enable rapid prototyping and R&D.

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Thank You!

Q&A

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